TANAY TIJARE

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TECHNICAL SKILLS

- **Programming & Frameworks:** Python, Object Oriented Programming, JAVA, FastAPI, SQLAlchemy
- **AI & LLM:** LLM APIs, Prompt Engineering, Structured Outputs, Pydantic/JSON Schema, Conversational AI, LLM-based Information Extraction, Prompt Chaining, ReAct/Tool Use, Hallucination Prevention, RAG, LangChain, LangGraph
- **APIs, Tools and Databases:** REST APIs, JWT Authentication & Authorization, Postman, PostgreSQL (Amazon RDS), Docker, Docker Compose, Jenkins, Nginx, Pandera
- **Data Analysis & ML:** Pandas, NumPy, Seaborn, Matplotlib, scikit-learn, TensorFlow, Keras, Data Preprocessing, Data Cleaning
- **Document Processing & Deployment:** PDF/DOCX Processing, JSON, Groq API, Render, Vercel

PROFESSIONAL EXPERIENCE

Associate Software Engineer, TCS, Nagpur 04/2025 | 06/2026

- Designed a **multi-service FastAPI backend**, splitting authentication and data validation into two independently deployable microservices communicating over REST.
- Implemented **JWT-based authentication and authorization** with bcrypt password hashing; integrated SQLAlchemy ORM with **PostgreSQL** for secure, efficient data management.
- Built a **dynamic CSV validation engine** using Pandera with configurable, database-stored validation rules, processing large files in chunks via Pandas to avoid memory overload.
- Configured **Nginx** as a reverse proxy across services and **containerized** the full stack using Docker and docker-compose for consistent deployment across environments.
- Set up a **Jenkins CI/CD pipeline** running in Docker to automate build and deployment workflows, reducing manual release effort.

PROJECTS

AI-Powered Portfolio & JD Matching Assistant | Python, FastAPI, Groq LLM, Pydantic, JavaScript

- Built and deployed a fully **production-ready LLM-powered portfolio assistant** that enables recruiters to interact with a candidate through a conversational chatbot and ask contextual questions about skills, experience, projects, and professional background.
- Developed an **AI resume parsing pipeline** that extracts information from PDF/DOCX resumes and converts unstructured resume content into structured candidate profiles using Groq LLMs, Pydantic models, JSON Schema, and structured JSON outputs.
- Designed **prompt-engineering workflows** with role definition, task constraints, source-of-truth rules, fallback handling, and hallucination prevention to keep responses grounded in verified candidate information.
- Implemented **conversation history and streaming LLM responses** using FastAPI and JavaScript to provide a ChatGPT-like conversational experience with incremental response rendering.
- Built an **AI-powered JD matching feature** that analyzes uploaded job descriptions against the candidate profile and generates match percentage, matched skills, missing skills, relevant experience/projects, and an explanation.
- Implemented **PDF/DOCX document processing**, temporary file cleanup, input validation, CORS configuration, environment-based API key management, and deployment cleanup for reliable production operation.
- Deployed the **FastAPI backend on Render** and **frontend on Vercel**, integrating both through REST APIs and validating the complete application flow in the deployed environment.
- **Live:** ai-portfolio-chatbot-sandy.vercel.app **GitHub:** github.com/tanay330/ai-portfolio-chatbot

Customer Segmentation And Purchase Prediction System | Python, Deep Learning, Flask, Streamlit, TensorFlow

- Developed an **AI-powered analytics system** to process CSV data for RFM analysis, churn prediction, customer segmentation, and market basket analysis to support targeted marketing strategies.
- Built and trained **deep learning models** using TensorFlow/Keras to identify churn-prone customers and segment users based on behavioral patterns.
- Performed **data preprocessing, feature engineering,** and exploratory data analysis using Pandas, NumPy, Matplotlib, and Seaborn to improve data quality and model performance.
- Created an interactive **web application** using Flask and Streamlit to enable data upload, visualization of insights, and access to predictive outputs.

EDUCATION

Priyadarshini College of Engineering, Nagpur *July 2020 - May 2024*
Bachelor of Technology, Computer Technology

Jeevan Vikas Vidyalaya, HSC *June 2018 - May 2020*
12th standard State Board

Jeevan Vikas Vidyalaya, SSC *June 2016 – June 2018*
10th standard State Board